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Rio de Janeiro, Brazil

Raul Pereira da Silva

Software Engineer

SUMMARY

Backend-focused software engineer with 20+ years of experience across product development, system integration, and technical leadership.

In the last 4 years, I've specialized in building backend services with Elixir, supporting web, mobile, and operational platforms at scale. I've also delivered an agentic AI system integrated with Elixir, combining the functional paradigm with modern AI capabilities.

Before that, I worked for many years with Java and Kotlin. Earlier in my career, I worked extensively on UI/UX-driven applications and led teams, projects, and system integrations.

I hold post-graduate degrees in "Systems Analysis, Design and Management" and in "Ergonomics, Usability and Human-Computer Interaction: Ergodesign and Interface Evaluation". I enjoy solving real problems, shipping reliable systems, and working with teams who care about quality and continuous improvement.

EXPERIENCE

R3M Tech — Rio de Janeiro, Brazil · Remote
Founder — *Jun 2012 - Present*

My own company, through which I provide software development services as a contractor and take on freelance projects for research centers, mid-sized companies and startups.

- Creation, maintenance, and enhancement of applications or services
- Development of prototype applications for research

Clients: Arionkoder, Coverflex, Fiocruz, Meu Patrocínio, Bioxxi · *Projects:* iSeatz (Arionkoder), ARCA (Fiocruz), Consuxx (Bioxxi)

Arionkoder — Massachusetts, United States · Remote
Software Engineer — *Oct 2025 - Oct 2026*

As a software engineer at iSeatz through Arionkoder, I worked on the team that owns the hotel search service for Amex's travel platform: aggregating availability, rates and promotions, ranking results, and personalizing offers for logged-in members. My main stack was Elixir and Phoenix.

- Internationalization: mapped what the supplier integration covered in each market and built workarounds for the gaps to meet delivery deadlines
- Reworked the Oban job that caches data to reduce search latency
- Improved observability of that job to catch data sync issues proactively
- Improved logs and error messages across the platform
- Implemented OpenTelemetry context propagation across child and sibling spans for end-to-end tracing
- Built new features, adjusted existing ones, fixed bugs and ran data extractions for analysis

Projects: iSeatz · *Stack:* Elixir, Phoenix, Oban, Redis, ETS, PostgreSQL, OpenTelemetry, Jaeger, Grafana, Sumo Logic, AWS, Docker

CloudWalk, Inc. — São Paulo, Brazil · Remote
Software Engineer — *Sep 2024 - Sep 2025*

As a software engineer at CloudWalk, I was part of an amazing team that built JIM, an AI agent for banking operations on InfinitePay in Brazil and the US, using Elixir and the BEAM.

- Standardized observability across the agent with Prometheus and OpenTelemetry
- Supported the expansion to the US market
- Built an encrypted upload system for customer images, with access restricted by design
- Improved database performance
- Improved the LiveView backoffice
- Implemented Kafka integrations and new service-to-service authentication methods
- Built agent tools in collaboration with other teams and contributed to the MCP integration we built

Projects: InfinitePay, JIM · *Stack:* Elixir, Phoenix, LiveView, Oban, Kafka, PostgreSQL, Neo4j, Redis, OpenAI, Gemini, LangChain, Vector Databases, MCP, Rust, OpenTelemetry, Prometheus, Grafana, Google Cloud, Kubernetes

Coverflex — Braga, Portugal · Remote
Backend Engineer — *Aug 2022 - Jul 2024*

As a backend engineer at Coverflex, an employee benefits platform in Portugal, Italy and Spain, I supported the web, mobile and backoffice applications using Elixir and Phoenix.

- Automated invoice processing for childcare services
- Implemented dynamic pricing adjustments for cards
- Introduced dedicated IBAN for all companies
- Participated in the Santander kick-off and developed exclusive features
- Expanded support for multiple insurances
- Streamlined the employee re-onboarding process
- Delivered improvements to the Apple marketplace
- Participated in the Spanish market kick-off
- Developed and optimized various bulk operations features

Stack: Elixir, Phoenix, LiveView, Oban, Commanded, PostgreSQL, Redis, ETS, React.js, TypeScript, AWS, Docker, Kubernetes

Jaya Tech São Paulo, Brazil · Remote

Tech Coordinator *May 2021 - Jul 2022*

As a tech coordinator at Jaya Tech, I supported multiple teams delivering microservices-based solutions for clients such as PayGo, Swap, Vagas, CERC and Mercado Livre.

- Mentored and developed new tech leads
- Hired and set up new teams
- Focused on each person's professional growth, aiming to make the teams fully autonomous

Projects: PayGo, Swap, Vagas, Mercado Livre, CERC · *Stack:* Kotlin, Java, Spring Boot, Elixir, Phoenix, Clojure, Go, Python, Node.js, PHP, .NET, Android, iOS, PostgreSQL, Google Cloud, Kubernetes, Docker

Tech Lead *Feb 2019 - Apr 2021*

As a tech lead at Jaya Tech, I led the team behind microservices-based solutions for C6 Bank's mobile apps and back-office systems, while also working as a backend developer and solutions analyst.

- Mentored each team member's professional growth, aiming to make the team fully autonomous
- Worked hands-on as a backend developer, mainly with Kotlin

Projects: C6 Bank · *Stack:* Kotlin, Java, Spring Boot, Spring Security, Javalin, Koin, GraphQL, PostgreSQL, MongoDB, Redis, TypeScript, React.js, AWS, Docker, Kubernetes, Ansible, Prometheus, Grafana

Backend Developer *Feb 2018 - Feb 2019*

As a consultant at Jaya Tech, I worked as a backend developer and solutions analyst on microservices-based solutions for mobile apps and back-office systems, for clients such as Guiabolso and C6 Bank.

Projects: C6 Bank, Guiabolso · *Stack:* Kotlin, Java, Groovy, Spring Boot, Javalin, Koin, Vaadin, Gradle, MySQL, PostgreSQL, MongoDB, Elixir, AWS, Docker, Ansible

Supply Midia Rio de Janeiro, Brazil · On-site

Fullstack Developer *Oct 2017 - Feb 2018*

As a fullstack developer at Supply Midia, I developed and maintained new modules for Total Bank, an EDI system that integrates companies with their banks, using Java, JSF/PrimeFaces and PostgreSQL. I also built the website for Total Card, a card transaction manager, using WordPress.

Projects: Total Card, Total Bank · *Stack:* Java, JSF, PrimeFaces, PostgreSQL, Maven, Jenkins, WordPress, PHP

Sonda Rio de Janeiro, Brazil · On-site

Fullstack Developer *Jun 2017 - Oct 2017*

As a fullstack developer at Sonda, I provided development services to BR Distribuidora, using Java on the backend and AngularJS on the frontend. I was responsible for the initial production release of a sales force portal intended for rollout across all of its gas stations.

Projects: BR Distribuidora · *Stack:* Java, JAX-RS, AngularJS, Node.js, WebSphere, Oracle, PL/SQL

Meu Patrocínio Rio de Janeiro, Brazil · On-site

Frontend Developer *May 2017 - Jun 2017*

As a frontend developer at Meu Patrocínio, I worked on the startup's mobile app for Android and iOS, using Angular, TypeScript and NativeScript.

Stack: Angular, TypeScript, NativeScript, Android SDK, iOS SDK

BRQ Rio de Janeiro, Brazil · On-site

Senior System Analyst *Mar 2012 - Apr 2017*

As a senior system analyst at BRQ, I kept working with Petrobras on desktop and web projects, which I had started at Atos, and supported the BPM team. After that contract ended, I helped deliver projects for Bradesco Seguros.

Projects: Bradesco Seguros, Petrobras · *Stack:* Java, Java Swing, Java Web Start, Grails, JPA, Hibernate, Spring, Spring Remoting, JDBC, iReport, Oracle BPM, FileNet P8, SAP, WebLogic, WebSphere, Oracle, PL/SQL, H2

EARLIER EXPERIENCE

Atos	Senior System Analyst · Rio de Janeiro, Brazil · On-site	Aug 2007 - Mar 2012
BRQ	Requirements Analyst · Rio de Janeiro, Brazil · On-site	Sep 2006 - Jul 2007
Tecgraf / PUC-Rio	Mid-level System Analyst · Rio de Janeiro, Brazil · On-site	Apr 2005 - Aug 2006
Drive	Junior System Analyst · Rio de Janeiro, Brazil · On-site	Sep 2000 - Feb 2005
Caixa Econômica Federal	Internship · Rio de Janeiro, Brazil · On-site	May 2000 - Aug 2000

EDUCATION

PUC-Rio	Postgraduate, Ergonomics, Usability and Human-Computer Interaction: Ergodesign and Interface Evaluation	Mar 2006 - Dec 2006
PUC-Rio	Postgraduate, Systems Analysis, Design and Management	Sep 2002 - Aug 2004
UniverCidade	Undergraduate, Technologist in Data Processing	Jan 1998 - Sep 2001

SKILLS

AI	LLM, LangChain, LangGraph, OpenAI, Gemini, CrewAI, Vector Databases
AI coding assistants	Claude Code, Codex, Gemini CLI, Cursor, GitHub Copilot, Aider
Programming	Elixir, Erlang, Kotlin, Java, JavaScript, TypeScript, Groovy, C++, SQL
Frameworks	Phoenix, LiveView, Oban, Spring Boot, Node.js, React.js, Grails
Data and messaging	PostgreSQL, Redis, ETS, Kafka, MongoDB, Neo4j, Oracle, MySQL
Observability and infra	OpenTelemetry, Prometheus, Grafana, Jaeger, Docker, Kubernetes, AWS, Google Cloud
Practices	Distributed Systems, Microservices, Event Sourcing, Functional Programming, Team Leadership, Coaching and Mentoring, Scrum, Kanban
Spoken languages	Portuguese (native), English (professional), Spanish (elementary)